JSPM's
Rajarshi Shahu College of Engineering, Tathawade, Pune
(An Autonomous Institute affiliated to Savitribai Phule Pune University)
Department of Computer Science and Business Systems

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Project Presentation

Project Title:

Loan Default Prediction using Random Forest

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Subject: Machine Learning

Dataset and Problem

Give Me Some Credit (Kaggle): who is in the data, what we predict, what had to be fixed

150,000
labelled borrowers

10
numeric predictors

6.68%
delinquent borrowers

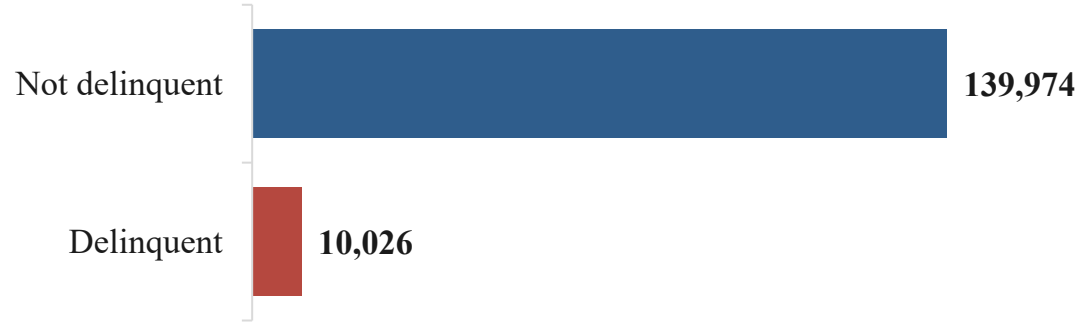
Ten predictors, in three groups

Payment history	Credit usage	Borrower profile
30–59 days late	Revolving utilisation	Age
60–89 days late	Debt ratio	Monthly income
90+ days late	Open credit lines	Dependents
	Real-estate loans	

Target: SeriousDlqin2yrs

1 = 90+ days past due within two years 0 = otherwise

Class balance: the delinquent class is rare



Always answering "no" scores 93.3% accuracy and catches no one.

Data issues found, and what was done

Issue	Affected	Handling
Monthly income missing	19.82%	Median + flag
Dependents missing	2.62%	Median + flag
Late counts coded 96 / 98	269 rows	Flag: 54.65% positive
Age equal to 0	1 row	Treated as missing
Outliers, duplicate rows	kept	Trees are rank-based

Methodology and Algorithm

A leakage-free pipeline around a Random Forest classifier

1 Load
150,000 labelled rows, 10 predictors, row index dropped

→

2 Clean + impute
96/98 codes and age < 18 become missing. Median fill plus missing flags

→

3 Split
Stratified 60 / 20 / 20: 90,000 train, 30,000 validation, 30,000 test

→

4 Compare
Five forests fitted on train, ranked by validation PR-AUC

→

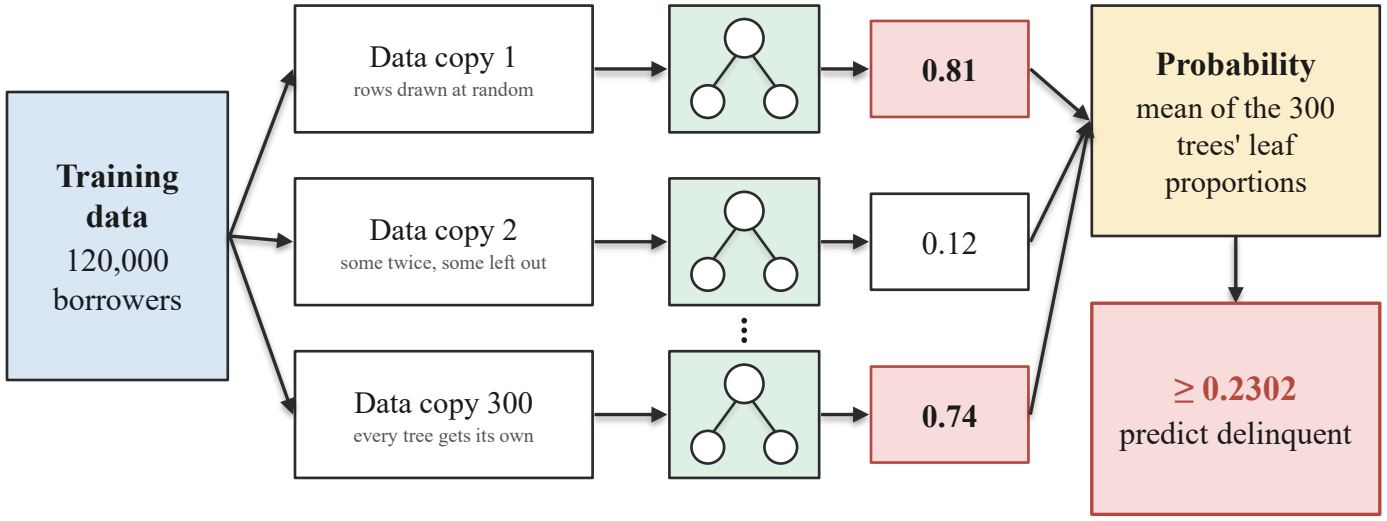
5 Set cut-off
Threshold 0.2302 chosen on validation by maximum F1

→

6 Test once
Refit on train + validation, then score the untouched test set

Every statistic is fitted on training data only. The test set is touched exactly once.

How the forest produces a prediction



Design choices, all decided on validation data

01 — 300 trees, minimum leaf size 20
Limits overfitting: validation PR-AUC 0.3820 against 0.3481 for unrestricted trees

02 — Class weighting tested, not used
PR-AUC stayed near 0.3776: it only slides along the same precision–recall curve

03 — Missing-value flags added
Best validation PR-AUC (0.3863). The 96/98 rows are 54.65% positive

04 — Cut-off 0.2302, not 0.50
Validation F1 0.4396 held up on the test set (0.4482)

Results and Evaluation

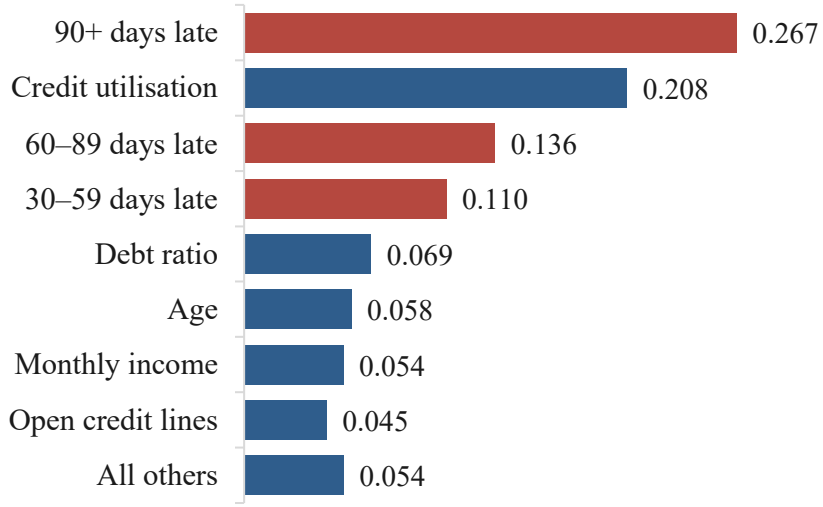
Held-out test set of 30,000 borrowers, scored exactly once

0.8654	0.4014	0.4868	0.4153	0.4482
ROC-AUC threshold-free	**PR-AUC** chance level 0.067	**Recall** caught 976 of 2,005	**Precision** at cut-off 0.2302	**F1-score** at cut-off 0.2302

Confusion matrix

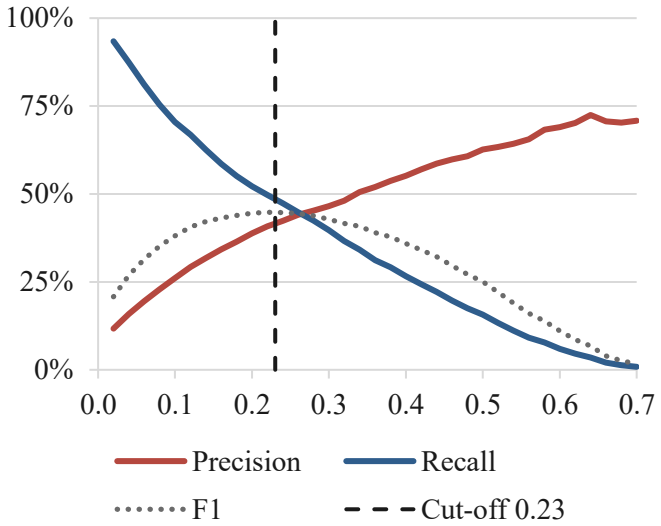
	Predicted: not delinquent	Predicted: delinquent
Actual not delinquent	26,621 true negatives	1,374 false alarms
Actual delinquent	1,029 missed	976 caught

What the forest relies on



Payment history (red) holds 51% of importance. Not causation.

Moving the cut-off



Lower cut-off: more caught, more false alarms.

Reading it: accuracy (92.0%) is below the 93.3% of always answering "no", so PR-AUC and recall carry the result. Five-fold cross-validation agrees: ROC-AUC 0.8632 ± 0.0053. **Limits:** 2011 US data, uncalibrated probabilities, no fairness test. An educational prototype.